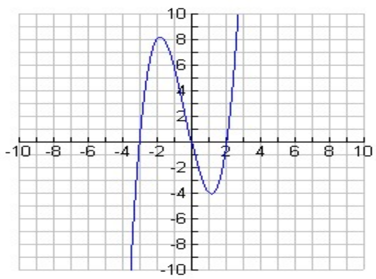
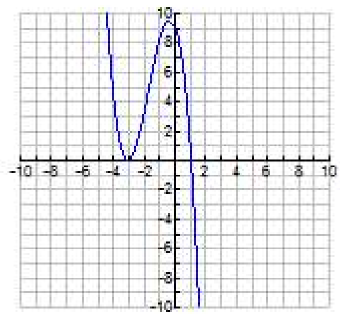
SOLUTIONS:



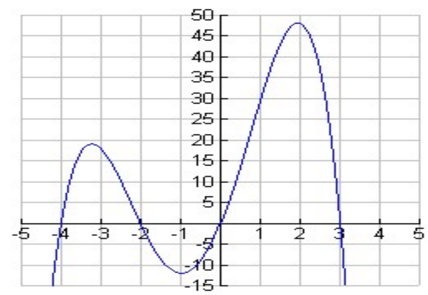
a) $y=x(x-2)(x+3)$

DEGREE: 3 (odd)
SIGN of the Lead Coefficient: positive
QUADRANTS: $3\leftrightarrow1$
ROOTS/X-intercepts: 0,2,-3
y-INTERCEPT: 0



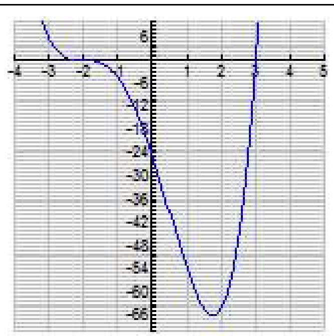
b) $y=-(x-1)(x+3)^2$

DEGREE: 3 (odd)
SIGN of the Lead Coefficient: negative
QUADRANTS: $2\leftrightarrow4$
ROOTS/X-intercepts: -3 (double root –bounce), 1
y-INTERCEPT: 9



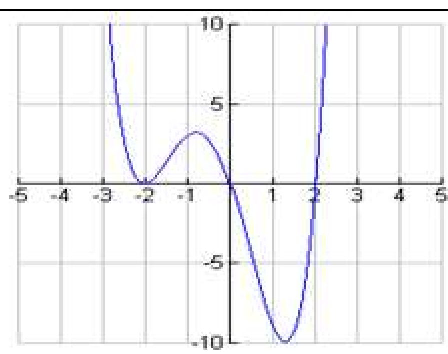
c) $y = -x(x-3)(x+2)(x+4)$

DEGREE: 4 (even)
SIGN: negative
QUADRANTS: $3\leftrightarrow4$
ROOTS: 0,3,-2,-4
y-INTERCEPT: 0



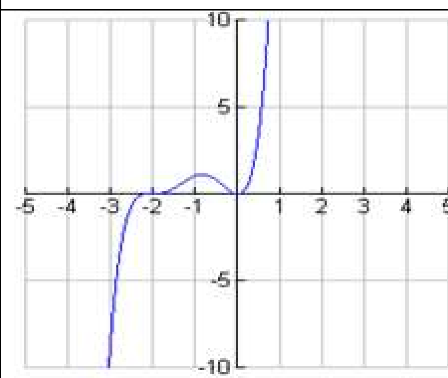
d) $y =(x+2)^3(x-3)$

DEGREE: 4 (even)
SIGN of the Lead Coefficient: positive
QUADRANT: $2\leftrightarrow1$
ROOTS/X-intercepts: -2 (saddle), 3
y-INTERCEPT: -24



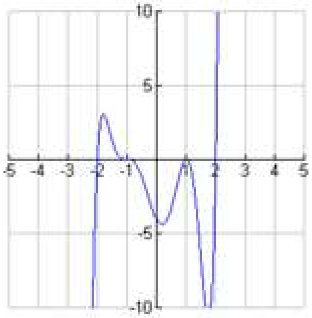
e) $y=x(x+2)^2(x-2)$

DEGREE: 4 (even)
SIGN: positive – opens up
QUADRANTS: $1\leftrightarrow2$
ROOTS: 0,2,-2(double root-bounce)
y-INTERCEPT: 0



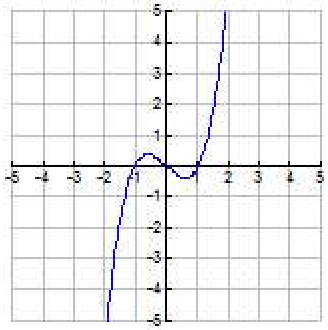
f) $y=x^2(x+2)^3$

DEGREE: 5 (odd)
SIGN: positive
QUADRANTS: $3\leftrightarrow1$
ROOTS: 0(double root—bounce), -2(saddle)
y-INTERCEPT: 0



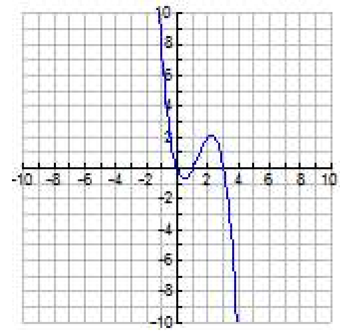
g) $y = (x-1)^2(x+1)^3(x-2)(x+2)$

DEGREE: 7 (odd)
SIGN: positive
QUADRANTS: 3↔1
ROOTS: 1(double root—bounce), -1(saddle),2, -2
y-INTERCEPT: -4



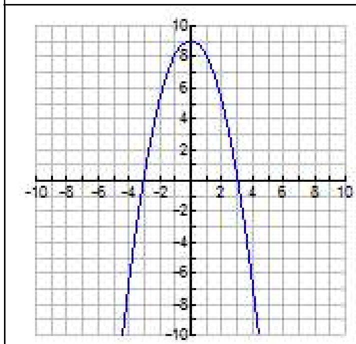
h) $y = x^3 - x$ (Fully factor this before graphing!)

DEGREE: 3 (odd)
SIGN: positive
QUADRANTS: 3↔1
ROOTS: -1, 0, 1
y-INTERCEPT: 0



i) $y = -x^3 + 4x^2 - 3x$ (Fully factor this before graphing!)

DEGREE: 3 (odd)
SIGN: negative
QUADRANTS: 2↔4
ROOTS: 0, 1,3
y-INTERCEPT: 0



j) $y = -(x^2 - 9)$ (Fully factor this before graphing!)

DEGREE: 2 (even)
SIGN: negative – opens down
QUADRANTS 3↔4
ROOTS/X-INTERCEPTS: -3,3
y-INTERCEPT: 9